



JIYOUN LEE

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Education

Dongduk Women's University

B.S. in Information Statistics

- Double Major in Computer Science
- Total GPA : 3.83/4.5 · Major GPA : 3.88/4.5

Seoul, S.Korea
Mar. 2021 – Feb. 2027 (Expected)

Johannes Gutenberg University Mainz

Exchange Semester (Computer Science)

- Relevant Coursework : Database Systems, Reinforcement Learning

Mainz, Germany
Mar. 2025 – Aug. 2025

Research Interests

- Recommender systems and personalization
- Natural language processing and large language models
- Trustworthy AI: reliability, uncertainty, and evaluation

Publications

- Lee, J. Y., Park, K. H., & Kim, H. H. "An Adaptive Cross-Domain Recommendation Framework for Structurally Disjoint Domains via Semantic Representation and Similar-User Knowledge Transfer" *Applied Sciences, under review (revision submitted)*.
- Lee, J. Y., Park, K. H., & Kim, H. H. "Cross-Domain Recommendation and Cold-Start Mitigation via Textual Semantic Mapping and Similar-User Transfer." *Proceedings of the Annual Symposium of KIPS 2026 (ASK 2026) Oral Presentation*.
- Ha, S. J., Oh, S. H., Ban, S. J., Lee, J. Y., & Kim, H. H. "A Factor Analysis and Regression-Based Prediction Model of Security Deposit Scam Amount for Preventing Rental Scam." *Proceedings of the Annual Symposium of KIPS 2024 (ASK 2024)*.

Project Experience

Adaptive Cross-Domain Recommendation for Structurally Disjoint Domains

First Author & Undergraduate Researcher

Seoul, S.Korea
Mar. 2026 – Aug. 2026

- Proposed a cross-domain framework with similar-user transfer for cold-start users.
- Designed domain-aware adaptive modules, improving Precision@10 by 0.175 (human) and 0.205 (LLM).
- Combined IRB-approved human and LLM-as-a-judge evaluation; analyzed human-LLM agreement.

Personalized Volunteer Activity Recommendation system

Team Leader

Seoul, S.Korea
Sep. 2025 – Dec. 2025

- Developed a system matching user hobby profiles with volunteer activities through semantic representations.
- Applied similar-user transfer to recommend activities to cold-start users with limited information.
- Received the Gold Prize and published a first-author conference paper at ASK 2026.

Generative AI Platform for Franchise Risk Prediction

Team Leader

Seoul, S.Korea
Sep. 2025 – Nov. 2025

- Developed an LLM-based conversational system for franchise risk analysis and explanation.
- Integrated LightGBM forecasting and SHAP-based interpretability.
- Led the team to the finals of the 13th Big Contest.

Personalized Fashion Recommendation Using Image and Survey Data

Team Leader

Seoul, S.Korea
Aug. 2024 – Nov. 2024

- Developed a recommendation pipeline combining item metadata, image features, and user survey responses.
- Applied object detection and ResNet18-based classification for fashion image preprocessing and style representation.
- Achieved 92.11% accuracy and a 90.37% F1-score and received an Excellence Award at the 2024 Data Creator Camp.

Individualized Housing Recommendation Service (3rd Open Data Forum)

Team member

Seoul, S.Korea

Oct. 2024 – Nov. 2024

Developed a personalized housing recommendation system using public data and user lifestyle features.

Designed a weighted ranking method and was selected as a finalist in the Open Data Forum.

Factor Analysis and Prediction of Security Deposit Guarantee Losses

Team member

Seoul, S.Korea

Sep. 2023 – May. 2024

Analyzed relationships among housing market indicators and security deposit guarantee incidents using public data.

Built linear and polynomial regression models to predict annual security deposit guarantee incident amounts.

Achieved an R^2 of 0.97 with the cubic regression model and published the findings at ASK 2024.

Strawberry Smart Farm Control Using Reinforcement Learning

Individual Project

Mainz, Germany

May. 2025 – Aug. 2025

Compared PPO and SAC in a custom continuous-control environment for temperature, humidity, and lighting.

Internships

Quad Miners (R&D Intern)

Seoul, S.Korea

Aug. 2026 – Feb. 2027 (Expected)

Supporting research on a government-funded AI security project for defense applications.

Conducting technical analysis and research documentation.

WPP Media Korea (Data Analysis Intern)

Seoul, S.Korea

Jan. 2026 – Feb. 2026

Analyzed advertising performance data for luxury-brand campaigns and identified key performance trends.

Provided data-driven recommendations for future campaign strategy and execution.

Awards & Honors

Awards

2026.01

Gold Prize, 2025-2 Capstone Design Collaborative Learning Competition

Dongduk Women's Univ

2025.01

Excellence Project Award, Outlier (Academic Club) Data Visualization Project Showcase

Dongduk Women's Univ

2024.12

Excellence Award, 2024 Data Creator Camp (DCC) Undergraduate Category

K-ICT Big Data Center

2024.06

Encouragement Award, The 2nd Generative AI Ideathon

Wrtn Technologies Inc

2024.01

Excellence Project Award, Outlier (Academic Club) Data Analysis Project Showcase

Dongduk Women's Univ

2023.08

Excellence Award, The 4th National University Student Sharing & Volunteer Hackathon

Korea Red Cross

Honors

2024

Semester High Honors, Awarded for outstanding academic performance (Fall Semester)

Dongduk Women's Univ

2024

Semester High Honors, Awarded for outstanding academic performance (Spring Semester)

Dongduk Women's Univ

2023

Semester High Honors, Awarded for outstanding academic performance (Spring Semester)

Dongduk Women's Univ

Extracurricular Activities

Outlier (Academic Data Society)

Seoul, S.Korea

Sep. 2023 – Jan. 2025

Data Analyst 6th & 7th

Led weekly data analysis projects involving predictive modeling, visualization, and research presentation.

BDA (Big Data Analysis Academic Society)

Seoul, S.Korea

Mar. 2024 – Sep. 2024

Data Pre-processing Track

Completed an intensive curriculum with weekly study sessions and hands-on assignments.

Specialized in data cleaning and feature engineering using Python to handle real-world datasets.

Teaching

CJ UNIT (11th Gen), CJ OliveNetworks

Seoul, S.Korea

Aug. 2024 – Dec. 2024

SW Education Volunteer

Conducted weekly software coding workshops for a full semester to bridge the digital divide for underserved students.

Mathematics Mentor

Seoul, S.Korea

Jul. 2023 – Sep. 2023

Ansan Youth Shelter

Provided personalized mathematics tutoring and academic mentoring for underprivileged youth at a residential shelter.

Certifications & Skills

Certificate

2024.09 - SQLD

Skills

Programming

Python, R, SQL, C, Java

Machine Learning

PyTorch, TensorFlow, scikit-learn, Sentence-Transformers, LLM APIs

Data Analysis

Pandas, NumPy, Matplotlib, Tableau